

1 Regular expressions

- An alternative way to describe regular languages
 - Common in scripting languages (perl, python, etc.) & UNIX utilities (awk, sed, grep, etc.)
 - Defined by the “regular operations”:
 - Union ($\cup$ or $+$)
 - Concatenation ($\cdot$ or nothing)
 - Star ($*$).
- * Note: $+$ is similar to star; except that it cannot include empty set: $R^+ = R^* - \{\emptyset\}$.

Examples Assume $\Sigma = \{0, 1\}$

- $0^*10^* = \{w \mid w \text{ contains exactly one } 1\}$
- $\Sigma^*1\Sigma^* = \{w \mid w \text{ contains at least one } 1\}$
- $\Sigma^*001\Sigma^* = \{w \mid w \text{ contains } 001 \text{ as a substring}\}$
- $(\Sigma\Sigma)^* = \{w \mid w \text{ contains an even number of characters}\}$
- $\Sigma\Sigma^* = \{w \mid w \neq \epsilon\}$
- $01 + 11 = \{01, 11\} = (0 + 1)1$
- $\emptyset^* = \{\epsilon\}$

Precedence of operators: 1. Star, 2. Concatenation, 3. Union

Formal Definition:

R is a *regular expression* if R is one of the following:

- $a, a \in \Sigma$
- ϵ
- $\emptyset$
- $R_1 + R_2$
- $R_1 \cdot R_2$
- R_1^*

where R_1 and R_2 are regular expressions.

1.1 Equivalence of REs and FAs

Theorem 1.1 *A language is regular iff it can be generated by some regular expression (RE).*

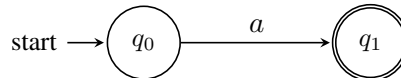
This will be proven in two parts:

Lemma 1.2 *Every language generated by a RE can be generated by some DFA/NFA.*

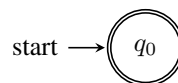
Lemma 1.3 *Every language generated by a DFA/NFA can be generated by some RE.*

Proof of Lemma 1.2. We will give an NFA for each of the 6 cases that define a RE.

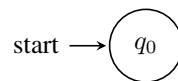
- $R = a$ for some $a \in \Sigma$



- $R = \epsilon$



- $R = \emptyset$

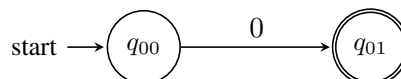


The NFA for the remainder can be constructed just in the same way as the NFA constructions we gave for the regular language operators:

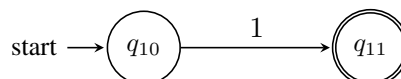
- $R_1 + R_2$
- $R_1 \cdot R_2$
- R_1^* ■

Examples

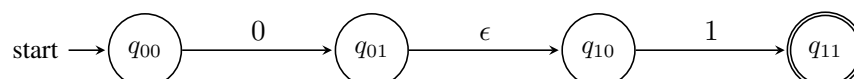
- 0:



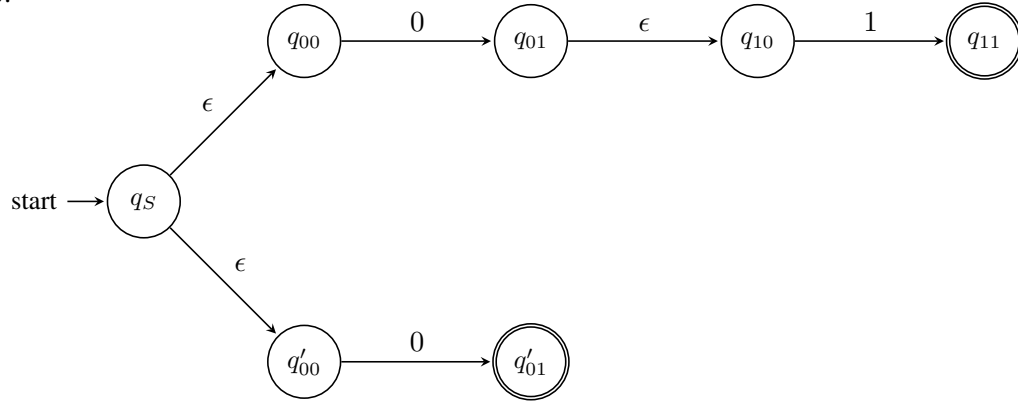
- 1:



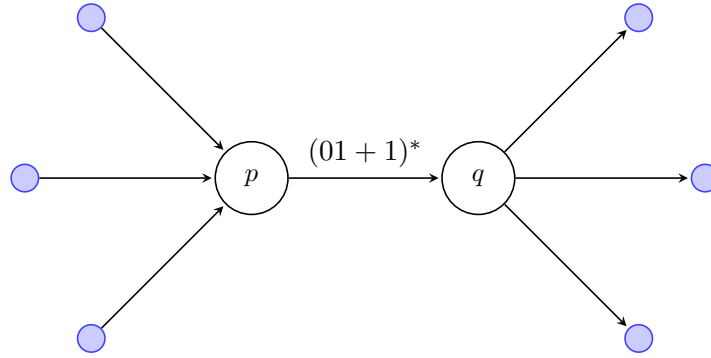
- 01:



- $01 + 0$:



For Lemma 1.3, we define Generalized NFA (GNFA), which takes regular expressions for its transitions:
E.g.,:

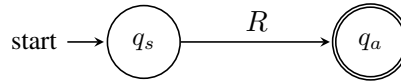


A GNFA also has the following properties:

- The start state does not have any incoming transitions.
- The accepting state (there is only one accepting state) does not have any outgoing transitions.
- Except for these, there is exactly one transition from every state to every other state.

Proof of Lemma 1.3. For this, we will:

1. Convert the given DFA into a GNFA (generalized NFA).
2. Reduce the number of states in the GNFS one by one until just 2 states remain.



which will be equivalent to a regular expression R .

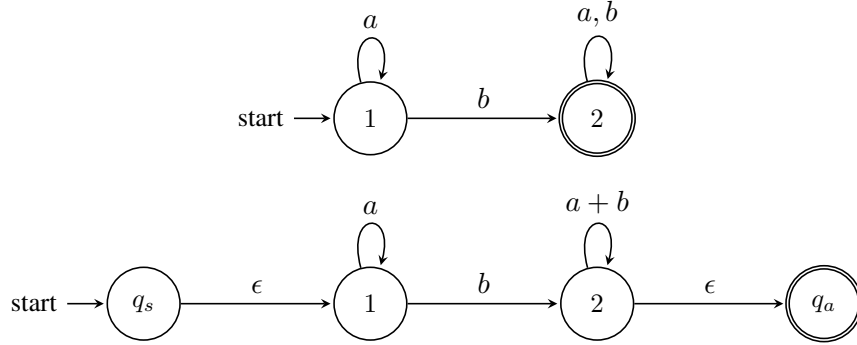
Definition: A GNFA is a 5-tuple $(Q, \Sigma, \delta, q_s, q_a)$:

- Q is a finite set of states.
- Σ is the alphabet.
- q_s is the start state.
- q_a is the accepting state.
- $\delta : (Q - \{q_a\}, Q - \{q_s\}) \rightarrow R$ is the transition function, where R denotes the set of regular expressions over Σ .

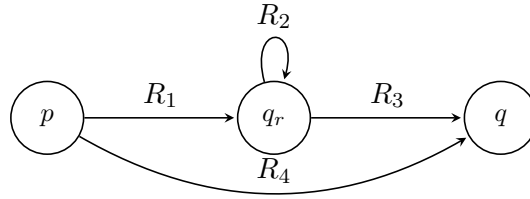
To convert a DFA to a GNFA:

- Add two new states q_s and q_a .
- Link q_s to the DFA's start state by ϵ . Link the DFA's accept state to q_a by ϵ .
- Assume an $\emptyset$ -transition for the missing arcs (just for the completeness of the GNFA). ■

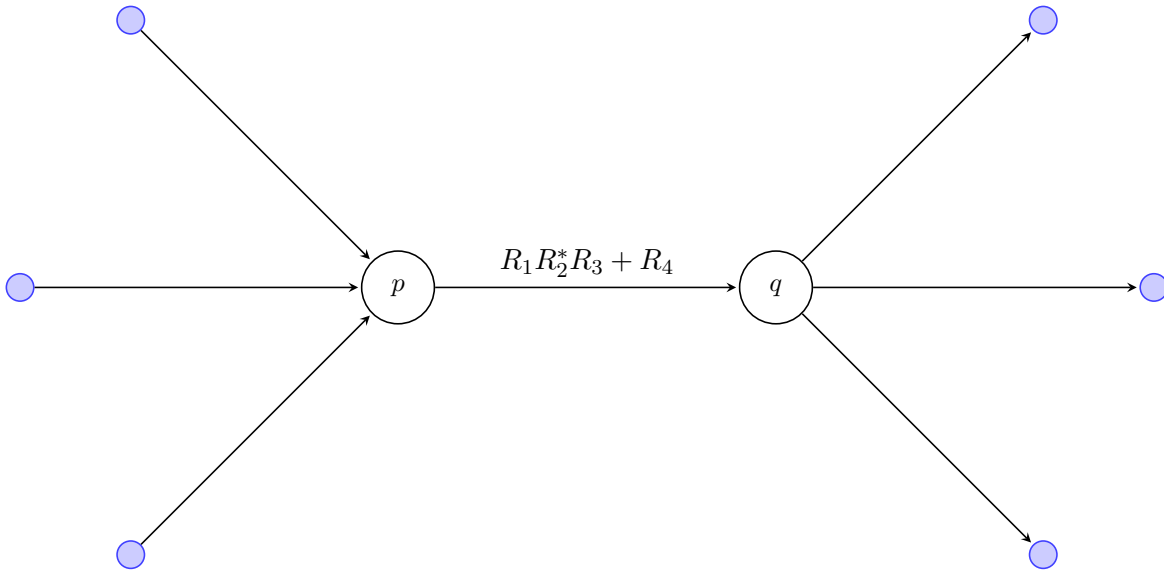
Examples Converting DFA.



We just assume $\emptyset$ -transitions; no need to write them. To reduce the GNFA by removing a state q_r :



update the transition function as:



and do this for all remaining states.

Reduce (G, q_r):

(Reduce GNFA G by removing state q_r)

Given $G = (Q, \Sigma, \delta, q_s, q_a)$, output $G = (Q', \Sigma, \delta', q_s, q_a)$ where:

$$Q' = Q - \{q_r\}$$

$$\delta'(p, q) = \delta(p, q_r)\delta(q_r, q_r)^*\delta(q_r, q) + \delta(p, q)$$

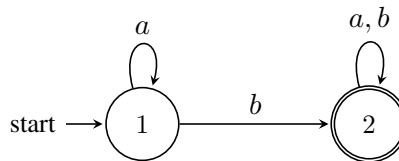
Claim: $L(G') = L(G)$

Algorithm 1 Convert(G)

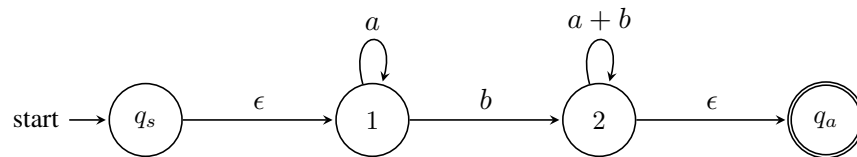
```
if  $|Q| = 2$  then
  return  $\delta(q_s, q_a)$ 
else
  choose any state  $q \in Q - \{q_s, q_a\}$ 
   $G' \leftarrow Reduce(G, q)$ 
  return  $Convert(G')$ 
end if
```

Examples Convert DFA into a RE

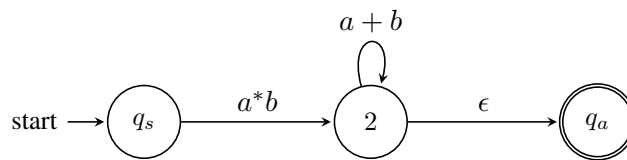
DFA:



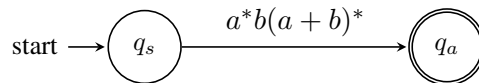
GNFA:



remove 1:

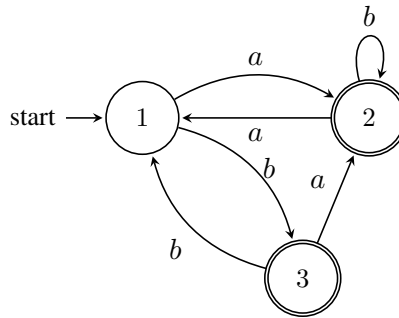


remove 2:

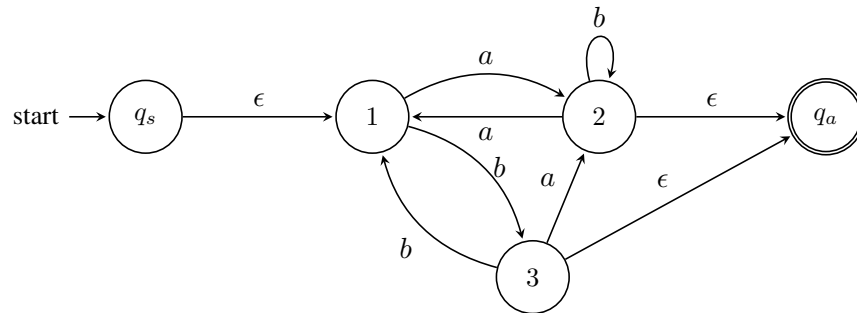


$$R = a^*b(a + b)^*$$

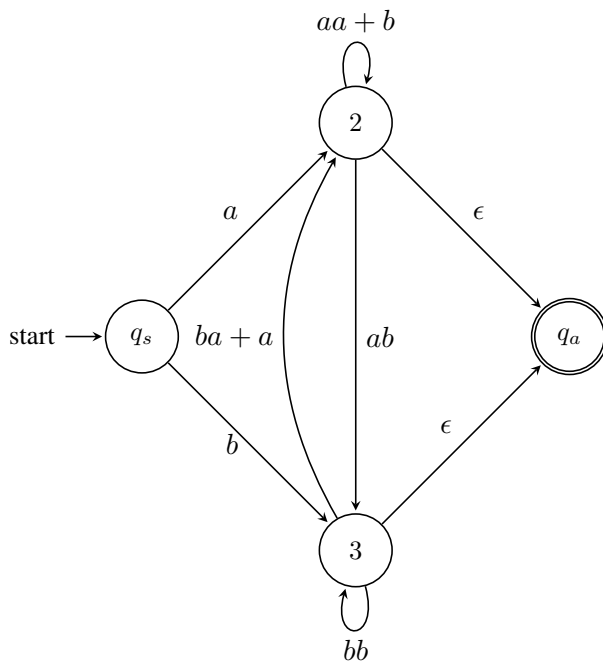
Examples Convert DFA into a RE
DFA:



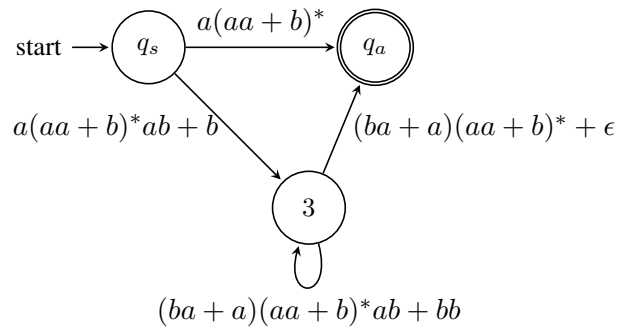
GNFA:



remove 1:



remove 2:



$$R = (a(aa + b)^*ab + b)((ba + a)(aa + b)^*ab + bb)^*((ba + a)(aa + b)^* + \epsilon) + a(aa + b)^*$$

Rule: $\delta'(p, q) = \delta(p, q_r)[\delta(q_r, q_r)]^* \delta(q_r, q) + \delta(p, q)$

Steps:

Remove 1:

$$\begin{aligned}\delta(q_s, 2) &= \delta(q_s, 1)[\delta(1, 1)]^* \delta(1, 2) + \delta(q_s, 2) \\ &= \epsilon a \\ &= a\end{aligned}$$

$$\begin{aligned}\delta(q_s, 3) &= \delta(q_s, 1)[\delta(1, 1)]^* \delta(1, 3) + \delta(q_s, 3) \\ &= \epsilon b \\ &= b\end{aligned}$$

$$\begin{aligned}\delta(2, 2) &= \delta(2, 1)[\delta(1, 1)]^* \delta(1, 2) + \delta(2, 2) \\ &= aa + b\end{aligned}$$

$$\begin{aligned}\delta(2, 3) &= \delta(2, 1)[\delta(1, 1)]^* \delta(1, 3) + \delta(2, 3) \\ &= ab\end{aligned}$$

$$\begin{aligned}\delta(3, 2) &= \delta(3, 1)[\delta(1, 1)]^* \delta(1, 2) + \delta(3, 2) \\ &= ba + a\end{aligned}$$

$$\begin{aligned}\delta(3, 3) &= \delta(3, 1)[\delta(1, 1)]^* \delta(1, 3) + \delta(3, 3) \\ &= bb\end{aligned}$$

Remove 2:

$$\begin{aligned}\delta(q_s, 3) &= \delta(q_s, 2)[\delta(2, 2)]^* \delta(2, 3) + \delta(q_s, 3) \\ &= a(aa + b)^* ab + b\end{aligned}$$

$$\begin{aligned}\delta(q_s, q_a) &= \delta(q_s, 2)[\delta(2, 2)]^* \delta(2, q_a) + \delta(q_s, q_a) \\ &= a(aa + b)^*\end{aligned}$$

$$\begin{aligned}\delta(3, 3) &= \delta(3, 2)[\delta(2, 2)]^* \delta(2, 3) + \delta(3, 3) \\ &= (ba + a)(aa + b)^* ab + bb\end{aligned}$$

$$\begin{aligned}\delta(3, q_a) &= \delta(3, 2)[\delta(2, 2)]^* \delta(2, q_a) + \delta(3, q_a) \\ &= (ba + a)(aa + b)^*\end{aligned}$$

Remove 3:

$$\begin{aligned}\delta(q_s, q_a) &= \delta(q_s, 3)[\delta(3, 3)]^* \delta(3, q_a) + \delta(q_s, q_a) \\ &= (a(aa + b)^* ab + b)((ba + a)(aa + b)^* ab + bb)^*((ba + a)(aa + b)^*) + a(aa + b)^*\end{aligned}$$

1.2 Pumping Lemma

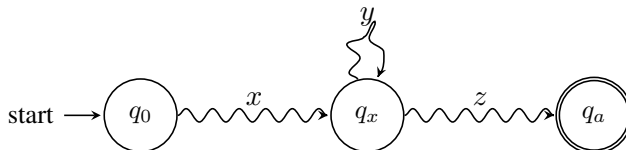
A simple but powerful idea to prove that a language is not regular (i.e., cannot be accepted by a DFA).

Consider a DFA M with n states. Then, for any string w where $|w| \geq n$ has to contain some cycle in its computation.

$$w = a_1 a_2 \dots a_m, m \geq n$$

$$q_0 \rightarrow q_1 \rightarrow \dots q_m \text{ has } m + 1 > n \text{ states on the path.}$$

Then by the pigeonhole principle, this path has some re-visited state:



$$xyz \in L$$

$$xy^2z \in L$$

...

$$xy^i z \in L \text{ for all } i \geq 0$$

$$\text{Note: } xy^0 z = xz \in L$$

Lemma 1.4 (Pumping Lemma)

If L is a RL, then there is some integer p ("pumping length") where if $w \in L$ and $|w| \geq p$, then w can be divided into three. $w = xyz$, such that:

1. $xy^iz \in L, \forall i \geq 0$
2. $|y| > 0$
3. $|xy| \leq p$

Proof of Lemma 1.4

If L is regular, it must be accepted by some DFA= $Q, \Sigma, \delta, q_0, F$. Let $p = |Q|$.

- The first two conditions follow from the previous argument.
- The third condition says that such a cycle can be found within the first p transitions. ■

To prove a language is regular: You are to prove that for **all sufficiently long strings** (w) in the language, there is **at least one way** (y) to generate new strings in the language by repeating the looping part any number (i) of times.

To prove a language is not regular: You are to find **at least one sufficiently long string** (w) in the language such that there is **no choice for any** y so that it's possible to generate new strings with all possible repetition (i).

Proving using Pumping Lemma:

	$w \in L, w \geq p$	y	i
language is regular	$\forall w \in L$	$\exists y$	$\forall i \geq 0$
language is NOT regular	$\exists w \in L$	$\nexists y$	$\exists i \geq 0$

Examples Show the following languages are not regular:

- $L_1 = \{0^n 1^n \mid n \geq 0\}$

Assume that L_1 is regular and let p be its pumping length. Consider $w = 0^p 1^p$.

- Since $|w| \geq p$, its DFA has to contain some cycle (i.e. $w = xyz$ such that $xy^iz \in L_1, \forall i \geq 0$).
- The cycle is within the first p characters because of the rule $|xy| \leq p$ (i.e., $y = 0^k$ for some $k > 0$), then

$$xy^2z = 0^{p+k}1^p \notin L_1$$

Therefore, L_1 is not regular. ■

Alternatively; consider $w = 0^{\frac{p}{2}} 1^{\frac{p}{2}}$.

- Since $|w| \geq p$, its DFA has to contain some cycle (i.e., $w = xyz$ such that $xy^iz \in L_1, \forall i \geq 0$). $|xy| \leq p$, therefore there are three possibilities for y :
- If the cycle is within the first $\frac{p}{2}$ characters (i.e., $y = 0^k$ for some $k > 0$), then

$$xy^2z = 0^{\frac{p}{2}+k}1^{\frac{p}{2}} \notin L_1$$

- Similarly, $y = 1^k$ is not possible:

$$xy^2z = 0^{\frac{p}{2}}1^{\frac{p}{2}+k} \notin L_1$$

- Finally, if y contains both 0 and 1 ($y = 0^k 1^l$; $k, l > 0$, then if we **pump up**, we get:

$$xy^2z = 0^{\frac{p}{2}-k}(0^k 1^l)^2 1^{\frac{p}{2}-l} = 0^{\frac{p}{2}} 1^l 0^k 1^{\frac{p}{2}} \notin L_1$$

That is, we will have some mixed string of 0s and 1s ($\notin L_1$). ■

- $L_2 = \{w \mid n_0(w) = n_1(w)\}$

Assume that L_2 is regular and let p be its pumping length. Consider $w = 0^p 1^p$.

- By the 3rd condition, w must contain a cycle within the first p characters.
- That is, if $y = 0^k$ for some $k > 0$,
- Then, $xy^2z = 0^{p+k}1^p \notin L_2$

Therefore, L_2 cannot be regular.

- $L_3 = \{w \mid w = w^R\}$ (i.e., w is a palindrome)

Assume that L_3 is regular and let p be its pumping length. Consider $w = 0^p 10^p$.

- We must have y among the first group of 0s; $y = 0^k$, $k > 0$.
- Then $xy^2z = 0^{p+k}10^p \notin L_3$

Hence, L_3 is not regular.

- $L_4 = \{w \mid w = vv^R, v \in \{0, 1\}^*\}$

Consider $w = 0^p 110^p$. Left as an exercise.

- $L_5 = \{w \mid w = vv, v \in \{0, 1\}^*\}$

Consider $w = 0^p 10^p 1$. $y = 0^k$ for some $k > 0$. $xy^2z = 0^{p+k}10^p 1 \notin L_5$.

- $L_6 = \{0^m 1^n \mid m \geq n\}$.

Consider $w = 0^p 1^p$. Then $y = 0^k$, $k > 0$.

“Pump down”: $xy^0z = xz = 0^{p-k}1^p \notin L_6$.

Also consider $w = 0^p 1^{p-1}$. Then $y = 0^k$, $k > 0$.

“Pump down”: $xz = 0^{p-k}1^{p-1}$. This is in L_6 for some k , and not in L_6 for other k .

- $L_7 = \{0^m 1^n \mid m \geq n\}$.

Alternatively, we can use the closure property: if L^R is not regular, then L is not regular.

$L_7^R = \{1^n 0^m \mid n \leq m\}$.

Consider $w = 1^p 0^p$. Then $y = 1^k$, $k > 0$.

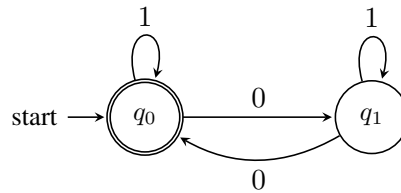
Pump up: $xy^2z = 1^{p+k}0^p \notin L_7^R$.

- $L_8 = \{w \mid n_0(w) \text{ is even}\}$.

Consider $w = 0^{2p}$. Then $y = 0^k$, $k > 0$.

Pump up $xy^2z = 0^{2p+k}$. If k is odd, $xy^2z \notin L_8$. Hence, L_8 is not regular.

⇒ **WRONG!** (Why?)



- $L_9 = \{0^m 1^n \mid m \neq n\}$.

Consider $L'_9 = \{0^m 1^n \mid m = n\}$.

$L'_9 = 0^* 1^* - L_9$. We know that L'_9 is not regular. If L_9 were regular, then by closure under “-”, L'_9 would be regular. Since L'_9 is not regular, L_9 is not regular.

- $L_{10} = \{w \mid n_0(w) \neq n_1(w)\}.$

Consider $\overline{L_{10}}$. Left as an exercise.

- $L_{11} = \{0^{n^2}, n \geq 1\} = \{0, 0000, 000000000, \dots\}.$

Consider $w = 0^{p^2}$. Then $y = 0^k$, $k > 0$.

Pump up: $xy^2z = 0^{p^2+k}$. Note that $p^2 + k$ cannot be a square. The next square is $(p+1)^2 = p^2 + 2p + 1 > p^2 + k$ given that $k < p$.

- $L_{12} = \{0^n \mid n \text{ is a prime number}\}.$

Assume L_{12} is regular and let p be its pumping length.

Consider $w = 0^q$ for some prime $q \geq p$. Then $y = 0^k$, $k > 0$.

$$\begin{aligned} xy^2z &= 0^{q+k} \\ xy^2yz &= 0^{q+2k} \\ &\vdots \\ &\vdots \\ xy^iz &= 0^{q+(i-1)k} \end{aligned}$$

Then for $i = q + 1$, we have $xy^{q+1}z = 0^{q+qk} = 0^{q(1+k)} \notin L_{12}$ for any $k > 0$ since $q(1+k)$ cannot be a prime number.

- $L_{13} = \{0^m 1^n \mid m \neq n\}.$

Consider $w = 0^p 1^{p+p!}$. Then $y = 0^k$, $1 \leq k \leq p$.

Then, $xy^iz = 0^{p+(i-1)k} 1^{p+p!}$, and for $i - 1 = \frac{p!}{k}$ (i.e. $i = \frac{p!}{k} + 1$).

We have $xy^iz = 0^{p+p!} 1^{p+p!} \notin L_{13}$.

- $L_{14} = \{w \mid w^R \neq w\}.$

Consider $w = 0^p 10^{p+p!}$. Then $y = 0^k$, $k > 0$.

$xy^iz = 0^{p+(i-1)k} 10^{p+p!}$, and for $i = \frac{p!}{k} + 1$, we have:

$0^{p+p!} 10^{p+p!} \notin L_{14}$.

- $L_{15} = \{0^m 1^n \mid \gcd(m, n) = 1\}$. (i.e. m and n are relatively prime)

For some prime $q \geq p + 2$, consider $w = 0^{q+(q-1)!} 1^{(q-1)!}$.

Then $y = 0^k$, $1 \leq k \leq p$.

Pump down: $0^{q-k+(q-1)!} 1^{(q-1)!} \notin L_{15}$. Note that $(q-k)$ is a non-trivial divisor for both sides.

1.3 Decision Questions about RLs

Given an n -state DFA, how can you tell whether its language is:

1.3.1 Finite/Infinite

Check all strings with length $n \leq |w| \leq 2n$. If any such $w \in L$, then L is infinite; otherwise, L is finite.

Theorem 1.5 *For an n -state DFA M , $L(M)$ is infinite iff $L(M)$ contains some string of length ℓ where $n \leq \ell \leq 2n$.*

Proof First, if $L(M)$ contains a string w of length $[n, 2n)$, by the pumping lemma, we can produce infinitely many strings in $L(M)$: $xy^iz, \forall i \geq 0$ for some partition in $w = xyz$.

Conversely, if $L(M)$ is infinite, then $\exists w', |w'| \geq n$. If $|w'| < 2n$, we are done (not possible). If $|w'| \geq 2n$, then from this w' we can obtain some w , $n \leq |w| < 2n$ by removing the cycles; since each basic cycle will have length $\leq n$. ■

1.3.2 Empty

Check all strings with length $|w| < n$. $L(M)$ is empty iff no such w is in $L(M)$.